Stat 231 — Statistics

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Chapter 1

Introduction to Statistical Sciences

1.1 What is statistics about?

Stat 230 was about *probability*: given a model, what data should we expect? Stat 231 turns this around. Given *data*, what can we learn about the process or population that produced it, and how sure can we be? Statistical science is the discipline of planning studies, collecting data, analysing it with models that account for variability, and drawing conclusions with an honest statement of uncertainty.

Definition 1.1 (Empirical study)**.** An *empirical study* is one in which knowledge is gained by observation or experiment, i.e. by collecting data, rather than purely by theory.

Examples: estimating the fraction of Ontario voters who support a party (a *sample survey*); measuring whether a new fertilizer increases crop yield (an *experimental study*); studying whether smoking is associated with lung cancer using hospital records (an *observational study*).

1.1.1 Units, populations and processes

Definition 1.2.

- A *unit* is an individual person, place or thing about which we take measurements.
- A *population* is a collection of units (finite and existing, e.g. all UW undergraduates in Fall 2019).
- A *process* is a mechanism that generates units over time (e.g. the stream of light bulbs produced by a factory, or calls arriving at a call centre). Populations of units generated by processes are typically unbounded/conceptual.

Definition 1.3 (Variates)**.** A *variate* is a characteristic of a unit, usually denoted by a letter such as x or y. Types of variates:

- **continuous**: measured on a continuous scale (height, time, weight);
- **discrete**: takes countably many (usually integer) values (number of accidents, number of defects);
- **categorical**: unit falls into one of a set of non-numerical categories (hair colour, blood type, program of study);

- **ordinal**: categorical with a natural order (strongly disagree, disagree, neutral, agree, strongly agree);
- **complex**: images, text, curves, sequences.

Definition 1.4 (Attribute)**.** An *attribute* of a population or process is a function of the variates over the whole population: e.g. the population mean height, the proportion of smokers, the correlation between two variates. Attributes are usually unknown; the goal of a study is to learn about them from a sample.

Remark. Statistical questions are often classified as *descriptive* (what is the value of an attribute?), *causative* (does changing x cause a change in y?), or *predictive* (what will y be for a new unit?).

1.2 Numerical data summaries

Suppose we have data $y_1, \dots, y_n$ on a single variate, with order statistics $y_{(1)} \le y_{(2)} \le \cdots \le y_{(n)}$.

1.2.1 Measures of location

Definition 1.5.

- *Sample mean*: $\bar{y} = \dfrac{1}{n}\sum_{i=1}^{n} y_i$.
- *Sample median* $\hat{m}$: the middle value, $y_{((n+1)/2)}$ if n is odd and $\frac{1}{2}\big(y_{(n/2)} + y_{(n/2+1)}\big)$ if n is even.
- *Sample mode*: the most frequent value (or the class with the highest frequency in a grouped summary).

The median is *robust*: a single huge outlier changes the mean a lot but the median hardly at all.

Definition 1.6 (Sample quantiles)**.** For $0 < p < 1$, the pth *sample quantile* $q(p)$ is a value such that roughly a fraction p of the data lie at or below it. One common convention (used in these notes): let $m = (n+1)p$. If m is an integer, $q(p) = y_{(m)}$; otherwise, with $k = \lfloor m \rfloor$, interpolate:

$$q(p) = y_{(k)} + (m - k)\big(y_{(k+1)} - y_{(k)}\big).$$

The *lower quartile* is $q(0.25)$, the median is $q(0.5)$ and the *upper quartile* is $q(0.75)$. Different software packages use slightly different conventions; for large n they agree closely.

1.2.2 Measures of variability (dispersion)

Definition 1.7.

- *Sample variance*: $s^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 = \dfrac{1}{n-1}\Big(\sum y_i^2 - n\bar{y}^2\Big)$, and *sample standard deviation* $s = \sqrt{s^2}$.
- *Range*: $y_{(n)} - y_{(1)}$.
- *Interquartile range*: $\mathrm{IQR} = q(0.75) - q(0.25)$.

The divisor $n - 1$ (rather than n) makes S^2 an unbiased estimator of σ^2 (see Chapter 4). The IQR, like the median, is robust to outliers.

1.2.3 Measures of shape

Definition 1.8.

- *Sample skewness*: $g_1 = \dfrac{\frac{1}{n}\sum(y_i-\bar{y})^3}{\left[\frac{1}{n}\sum(y_i-\bar{y})^2\right]^{3/2}}$. Positive skewness means a long right tail; negative means a long left tail; $g_1 \approx 0$ for symmetric data.
- *Sample kurtosis*: $g_2 = \dfrac{\frac{1}{n}\sum(y_i-\bar{y})^4}{\left[\frac{1}{n}\sum(y_i-\bar{y})^2\right]^2}$. For Gaussian data $g_2 \approx 3$; $g_2 > 3$ indicates heavier tails (more extreme values) than the Gaussian, $g_2 < 3$ lighter tails (e.g. uniform data has kurtosis 1.8).

1.2.4 Bivariate summaries

For paired data (x_i, y_i), the *sample correlation* is

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}, \quad S_{xx} = \sum(x_i-\bar{x})^2,\ S_{yy} = \sum(y_i-\bar{y})^2,\ S_{xy} = \sum(x_i-\bar{x})(y_i-\bar{y}).$$

Always $-1 \le r \le 1$; r measures the strength of *linear* association only. For two categorical variates, a *two-way (contingency) table* of counts is the natural summary (Chapter 7).

Example 1.9 (Complete set of numerical summaries)**.** The waiting times (minutes) of $n = 12$ customers are

$$3.2,\ 4.1,\ 4.5,\ 5.0,\ 5.3,\ 5.8,\ 6.1,\ 6.4,\ 7.0,\ 7.9,\ 9.6,\ 12.3$$

(already sorted). Then $\sum y_i = 77.2$, $\sum y_i^2 = 567.06$, so

$$\bar{y} = \frac{77.2}{12} = 6.4333, \qquad S_{yy} = 567.06 - \frac{77.2^2}{12} = 567.06 - 496.6533 = 70.4067,$$

$$s^2 = \frac{70.4067}{11} = 6.4006, \qquad s = 2.5299.$$

Quantiles. For $p = 0.25$: $m = 13(0.25) = 3.25$, so $q(0.25) = y_{(3)} + 0.25(y_{(4)} - y_{(3)}) = 4.5 + 0.25(0.5) = 4.625$. For $p = 0.5$: $m = 6.5$, so $\hat{m} = (5.8 + 6.1)/2 = 5.95$. For $p = 0.75$: $m = 9.75$, so $q(0.75) = 7.0 + 0.75(0.9) = 7.675$. Hence IQR $= 7.675 - 4.625 = 3.05$, and the range is $12.3 - 3.2 = 9.1$.

Shape. With $m_k = \frac{1}{n}\sum(y_i-\bar{y})^k$ we get $m_2 = 5.8672$, $m_3 = 14.882$, $m_4 = 120.74$, so

$$g_1 = \frac{14.882}{5.8672^{3/2}} = 1.047, \qquad g_2 = \frac{120.74}{5.8672^2} = 3.51.$$

The data are right-skewed (mean 6.43 > median 5.95, $g_1 > 0$) with slightly heavier tails than a Gaussian. This is typical of waiting times.

1.3 Graphical data summaries

1.3.1 Histograms

Partition the range into intervals (bins) $[a_0, a_1), [a_1, a_2), \ldots, [a_{k-1}, a_k)$ and count the frequency f_j in each. A *relative frequency histogram* draws a rectangle over bin j with *area* f_j/n, i.e. height $f_j/(n(a_j - a_{j-1}))$. Then the total area is 1 and the histogram can be compared directly with a probability density function. With unequal bin widths, using heights = frequencies is a classic mistake: wide bins look too important.

1.3.2 Empirical cumulative distribution function

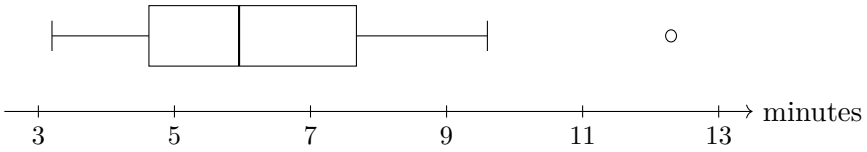
Definition 1.10 (ECDF)**.** $\hat{F}(y) = \dfrac{\#\{i : y_i \le y\}}{n}$, a step function jumping by $1/n$ at each observation. It estimates the population c.d.f. $F(y) = \mathrm{P}(Y \le y)$.

The ECDF uses all the data (no binning choice) and quantiles can be read off it. To check a model, plot $\hat{F}$ together with the fitted model c.d.f.

1.3.3 Boxplots

A *boxplot* draws a box from $q(0.25)$ to $q(0.75)$ with a line at the median. Whiskers extend to the most extreme observations within 1.5 IQR of the box; points beyond that are plotted individually as possible outliers. Side-by-side boxplots are excellent for comparing groups.

Example 1.11 (Boxplot of the waiting times)**.** In Example 1.9, $1.5\,\mathrm{IQR} = 4.575$, so the fences are $4.625 - 4.575 = 0.05$ and $7.675 + 4.575 = 12.25$. The value 12.3 exceeds the upper fence, so it is plotted as an outlier and the upper whisker stops at 9.6; the lower whisker goes to 3.2.



The longer right whisker and the outlier both display the right skewness.

1.3.4 Scatterplots and run charts

For bivariate data (x_i, y_i) a *scatterplot* shows the form (linear, curved), direction and strength of a relationship and reveals outliers. A *run chart* plots a variate against time order to show trends or shifts in a process.

1.3.5 Gaussian QQ plots

Definition 1.12 (QQ plot)**.** To check whether data look like a sample from a Gaussian distribution, plot the ordered data $y_{(i)}$ against the theoretical standard normal quantiles z_i with $\Phi(z_i) = \frac{i}{n+1}$ (or a similar plotting position). If the data are Gaussian, $y_{(i)} \approx \mu + \sigma z_i$, so the points should lie close to a straight line.

How to read a QQ plot:

- points on a straight line: Gaussian model reasonable (slope $\approx \sigma$, intercept $\approx \mu$);
- concave-up ("U") shape: right skew; concave-down: left skew;
- an "S" shape with both ends pulled away from the line: tails heavier than Gaussian (kurtosis > 3); the reverse S: lighter tails;
- some wiggle at the very ends is normal for small n, so don't over-interpret.

Common pitfalls

- The sample variance divides by $n-1$, not n.
- Correlation measures *linear* association: $r \approx 0$ does not mean "no relationship" (e.g. $y = x^2$ on symmetric x), and large $|r|$ does not imply causation (Chapter 8).
- Histogram areas, not heights, represent relative frequency.
- Always plot the data before computing summaries: means and standard deviations can hide outliers, skewness and multiple modes.

Summary

- Empirical studies collect data on *units* from a *population* or *process*; we measure *variates* and want to learn about *attributes*.
- Location: mean, median, mode. Spread: variance/standard deviation, range, IQR. Shape: skewness (0 for symmetric) and kurtosis (3 for Gaussian).
- Graphs: histograms (area = relative frequency), ECDF, boxplots (1.5 IQR rule), scatterplots, and Gaussian QQ plots (straight line means Gaussian).

Chapter 2

Statistical Models and Maximum Likelihood Estimation

2.1 Statistical models

Data vary from unit to unit and from sample to sample. A *statistical model* describes this variability with a probability distribution: we regard the observed data $y_1, \ldots, y_n$ as realizations of random variables $Y_1, \ldots, Y_n$ whose distribution depends on unknown *parameters* θ (possibly a vector). Parameters are linked to the attributes we care about (e.g. θ = proportion of voters supporting a party).

Remark (Notation). Capital letters Y are random variables; lower case y are observed values. An *estimate* $\hat{\theta}$ is a number computed from y; the corresponding *estimator* $\tilde{\theta}$ is the same function applied to the random Y.

2.1.1 Choosing a model

Model choice is guided by

- **background knowledge** of the process: a count of successes in n independent trials suggests Binomial(n, θ); counts of rare events in time or space suggest Poisson(θ); waiting times between events in a Poisson process suggest Exponential(θ); measurements that are sums of many small effects suggest G(μ, σ);
- **past experience** with similar data;
- **the data themselves**: histograms, ECDFs and QQ plots (Chapter 1).

"All models are wrong, but some are useful": a model is an approximation that should be good enough for the question at hand. In these notes G(μ, σ) denotes the Gaussian (normal) distribution with mean μ and *standard deviation* σ.

Model	p.f./p.d.f.	Mean	Variance
Binomial(n, θ)	$\binom{n}{y}\theta^y(1-\theta)^{n-y}$, $y = 0, \ldots, n$	$n\theta$	$n\theta(1-\theta)$
Poisson(θ)	$\theta^y e^{-\theta}/y!$, $y = 0, 1, \ldots$	θ	θ
Exponential(θ)	$\frac{1}{\theta}e^{-y/\theta}$, $y > 0$	θ	θ^2
G(μ, σ)	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(y-\mu)^2/(2\sigma^2)}$	μ	σ^2

2.2 The likelihood function

Definition 2.1 (Likelihood)**.** Suppose the data y have probability (density) function $f(y;\theta)$. The *likelihood function* is

$$L(\theta) = L(\theta; y) = c \cdot \mathrm{P}(\text{observing the data } y;\ \theta), \qquad \theta \in \Omega,$$

viewed as a function of θ with y fixed; c is any positive constant not depending on θ. For independent observations $y_1, \ldots, y_n$, $L(\theta) = \prod_{i=1}^{n} f(y_i;\theta)$. For continuous data we use the p.d.f. (which is proportional to the probability of observing a value in a small interval around y_i).

Definition 2.2 (MLE, log-likelihood, relative likelihood)**.**

- The *maximum likelihood estimate* (MLE) $\hat\theta$ is the value of θ that maximizes $L(\theta)$: the parameter value that makes the observed data most probable.
- The *log-likelihood* is $\ell(\theta) = \log L(\theta)$ (natural log). Since log is increasing, ℓ and L have the same maximizer, and ℓ turns products into sums. Usually we solve the *score equation* $\ell'(\theta) = 0$ and check it is a maximum.
- The *relative likelihood* is $R(\theta) = L(\theta)/L(\hat\theta)$, so $0 \le R(\theta) \le 1$ and $R(\hat\theta) = 1$. It measures how plausible θ is compared with the best value, and is free of the arbitrary constant c.

Remark. If two independent studies give data $y^{(1)}$ and $y^{(2)}$ about the same θ, the combined likelihood is the product $L_1(\theta)L_2(\theta)$ (log-likelihoods add).

2.3 MLEs for standard models

Theorem 2.3 (Binomial)**.** *If $Y \sim \mathrm{Binomial}(n,\theta)$ and we observe y, then $\hat\theta = y/n$.*

Proof. $L(\theta) = \theta^y(1-\theta)^{n-y}$ (dropping $\binom{n}{y}$), so $\ell(\theta) = y\log\theta + (n-y)\log(1-\theta)$ and $\ell'(\theta) = \frac{y}{\theta} - \frac{n-y}{1-\theta}$. Setting this to 0 gives $y(1-\theta) = (n-y)\theta$, i.e. $\theta = y/n$. Since $\ell''(\theta) = -y/\theta^2 - (n-y)/(1-\theta)^2 < 0$, it is a maximum (for $0 < y < n$; the boundary cases give $\hat\theta = 0$ or 1 directly). □

Theorem 2.4 (Poisson)**.** *If $Y_1, \ldots, Y_n$ are independent* Poisson(θ)*, then $\hat\theta = \bar y$.*

Proof. $L(\theta) = \prod \theta^{y_i} e^{-\theta}/y_i! \propto \theta^{\sum y_i} e^{-n\theta}$, so $\ell(\theta) = (\sum y_i)\log\theta - n\theta$ and $\ell'(\theta) = \sum y_i/\theta - n = 0$ gives $\hat\theta = \bar y$; $\ell'' < 0$. □

Theorem 2.5 (Exponential)**.** *If $Y_1, \ldots, Y_n$ are independent* Exponential(θ) *(mean θ), then $\hat\theta = \bar y$.*

Proof. $L(\theta) = \theta^{-n} e^{-\sum y_i/\theta}$, $\ell(\theta) = -n\log\theta - \sum y_i/\theta$, $\ell'(\theta) = -n/\theta + \sum y_i/\theta^2 = 0 \iff \theta = \bar y$. □

Theorem 2.6 (Gaussian)**.** *If $Y_1, \ldots, Y_n$ are independent* $\mathrm{G}(\mu,\sigma)$ *with both parameters unknown, then*

$$\hat\mu = \bar y, \qquad \hat\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar y)^2.$$

Proof. $\ell(\mu,\sigma) = -n\log\sigma - \frac{1}{2\sigma^2}\sum(y_i-\mu)^2$ (plus a constant). Then $\partial\ell/\partial\mu = \frac{1}{\sigma^2}\sum(y_i-\mu) = 0$ gives $\hat\mu = \bar y$ for every σ, and $\partial\ell/\partial\sigma = -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum(y_i-\mu)^2 = 0$ gives $\hat\sigma^2 = \frac{1}{n}\sum(y_i-\bar y)^2$. □

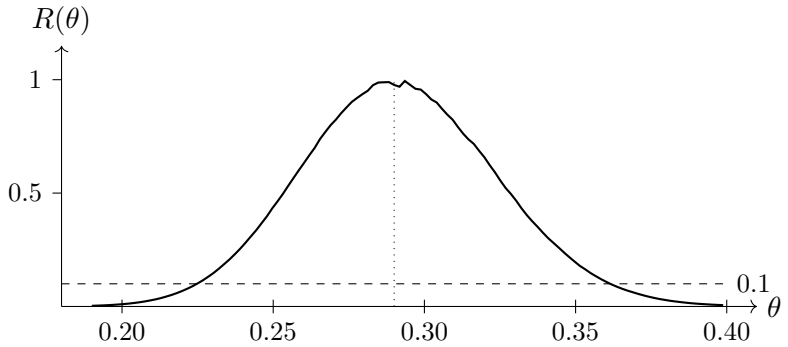
Note that the Gaussian MLE of σ^2 divides by n, whereas the sample variance s^2 divides by $n-1$. For large n the difference is negligible.

2.4 Worked examples

Example 2.7 (Binomial survey)**.** In a random sample of $n = 200$ students, $y = 58$ say they work part time. Model: $Y \sim \text{Binomial}(200, \theta)$. Then $\hat\theta = 58/200 = 0.29$ and

$$\ell(\theta) = 58 \log \theta + 142 \log(1-\theta), \qquad R(\theta) = \left(\frac{\theta}{0.29}\right)^{58} \left(\frac{1-\theta}{0.71}\right)^{142}.$$

For instance $R(0.25) = 0.438$ and $R(0.35) = 0.196$: $\theta = 0.25$ is quite plausible, $\theta = 0.35$ less so. Solving $R(\theta) = 0.1$ numerically gives $\theta \approx 0.225$ and 0.362 (this will be a *10% likelihood interval*, Chapter 4).



Example 2.8 (Poisson counts and invariance)**.** The numbers of help-desk calls in 10 randomly chosen five-minute periods were $3, 5, 2, 4, 6, 1, 3, 4, 2, 5$. Assuming independent Poisson(θ) counts, $\sum y_i = 35$ and $\hat\theta = 35/10 = 3.5$ calls per period. The relative likelihood is $R(\theta) = (\theta/3.5)^{35} e^{10(3.5-\theta)}$.

Theorem 2.9 (Invariance property)**.** *If $\hat\theta$ is the MLE of θ and $\tau = h(\theta)$, then the MLE of τ is $\hat\tau = h(\hat\theta)$.*

Proof sketch. For one-to-one h, L expressed in terms of τ is $L(h^{-1}(\tau))$, which is maximized when $h^{-1}(\tau) = \hat\theta$. For general h one defines the induced likelihood $L^*(\tau) = \sup_{\theta : h(\theta) = \tau} L(\theta)$, which is maximized at $\tau = h(\hat\theta)$. □

Example 2.10 (Using invariance)**.**

- Poisson calls: the MLE of $\mathrm{P}(Y = 0) = e^{-\theta}$ is $e^{-3.5} = 0.0302$, and of $\mathrm{P}(Y \le 1) = e^{-\theta}(1+\theta)$ is $e^{-3.5}(4.5) = 0.1359$.
- Exponential lifetimes (in years) of 8 components: $2.1, 0.4, 5.3, 1.8, 3.6, 0.9, 7.2, 2.7$. Then $\sum y_i = 24.0$, $\hat\theta = 3.0$. The MLE of $\mathrm{P}(Y > 5) = e^{-5/\theta}$ is $e^{-5/3} = 0.1889$, and of the median lifetime $\theta \log 2$ is $3 \log 2 = 2.079$ years.
- Gaussian: from $n = 10$ fill weights (grams) $12.4, 11.8, 13.1, 12.9, 12.0, 13.5, 12.7, 11.6, 12.2, 13.0$ we get $\sum y_i = 125.2$, $\sum y_i^2 = 1570.96$, so $\hat\mu = 12.52$, $S_{yy} = 1570.96 - 125.2^2/10 = 3.456$, $\hat\sigma^2 = 0.3456$, and by invariance $\hat\sigma = \sqrt{0.3456} = 0.5879$. (By contrast, $s = \sqrt{3.456/9} = 0.6197$.)

2.5 Checking the model

After fitting, ask whether the model is consistent with the data:

1. **Observed vs expected frequencies.** For discrete data (or grouped continuous data), compare the observed count in each class with the expected count $n\,\hat p_j$ computed from the fitted model. (Chapter 7 turns this into a formal test.)

2. **Graphs.** Superimpose the fitted p.d.f. on a relative frequency histogram, or the fitted c.d.f. on the ECDF; use a Gaussian QQ plot for Gaussian models.

3. **Summary statistics.** E.g. for a Poisson model the sample mean and sample variance should be similar; for an exponential model the sample mean and sample standard deviation should be similar; for a Gaussian model skewness ≈ 0 and kurtosis ≈ 3.

Example 2.11 (Observed vs expected under a Poisson fit)**.** In 200 one-minute intervals the numbers of cars passing a point were recorded:

cars	0	1	2	3	≥ 4
observed	38	80	50	20	12

The 12 intervals in the last class had 9 fours and 3 fives, so the total count is $80 + 2(50) + 3(20) + 4(9) + 5(3) = 291$ and $\hat{\theta} = 291/200 = 1.455$. The fitted probabilities $e^{-1.455}1.455^y/y!$ for $y = 0, 1, 2, 3$ are $0.2334, 0.3396, 0.2471, 0.1198$, and $\mathrm{P}(Y \geq 4) = 0.0601$. Expected frequencies (multiply by 200): $46.68, 67.92, 49.41, 23.96, 12.02$. The agreement is reasonable, with somewhat fewer zeros and more ones than the model predicts; Chapter 7 shows that this discrepancy is well within chance variation.

Common pitfalls

- The likelihood is a function of θ, not a probability distribution for θ: it need not integrate to 1.
- Do not forget to check that a stationary point is a maximum, and watch for boundary maxima (e.g. $y = 0$ in the binomial).
- Parametrization matters for writing formulas: our exponential has *mean* θ (rate $1/\theta$), and $\mathrm{G}(\mu, \sigma)$ uses the standard deviation.

Summary

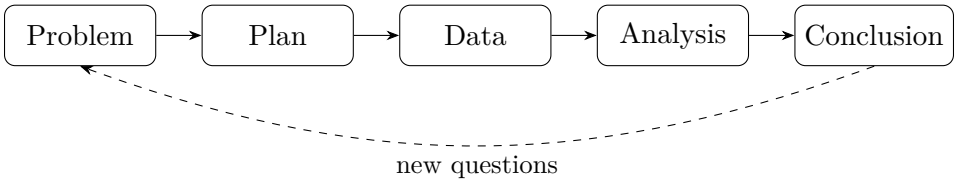
- $L(\theta) \propto \mathrm{P}(\text{data}; \theta)$; $\ell = \log L$; $R(\theta) = L(\theta)/L(\hat{\theta}) \in [0, 1]$.
- MLEs: binomial y/n; Poisson $\bar{y}$; exponential $\bar{y}$; Gaussian $\hat{\mu} = \bar{y}$, $\hat{\sigma}^2 = \frac{1}{n}\sum(y_i - \bar{y})^2$.
- Invariance: the MLE of $h(\theta)$ is $h(\hat{\theta})$.
- Check fit with observed vs expected frequencies, plots and summary statistics.

Chapter 3

Planning and Conducting Empirical Studies

3.1 The PPDAC framework

Stat 231 organizes every empirical study into five stages, abbreviated **PPDAC**: **P**roblem, **P**lan, **D**ata, **A**nalysis, **C**onclusion. The framework is a checklist for designing a study and for critically reading someone else's.



3.1.1 Problem

A clear statement of what we want to learn.

- **Target population/process**: the collection of units to which we want the conclusions to apply.
- **Variates**: response variate(s) y and explanatory variate(s) x.
- **Attributes** of interest: e.g. the mean of y, a proportion, a difference of means between two groups.
- **Type of question**: descriptive, causative or predictive.

3.1.2 Plan

How the data will be collected.

- **Study population**: the collection of units that *could* be selected in the study. Often different from the target population (e.g. target: all Canadian adults; study population: adults with a listed phone number; or target: humans, study population: laboratory mice).
- **Sampling protocol**: how units are selected from the study population, and the sample size n.

- **Measurement system**: how the variates are measured (instruments, questionnaire wording, who measures, when).
- For experiments: how the explanatory variates are assigned to units.

3.1.3 Data

Collect the data according to the plan, recording any departures from it (non-response, lost units, measurement problems). Data should be checked and cleaned and stored in a format suitable for analysis.

3.1.4 Analysis

Numerical and graphical summaries (Chapter 1), choice and fitting of a model (Chapter 2), estimation and testing (Chapters 4–7), and model checking.

3.1.5 Conclusion

Answer the question posed in the Problem step, in plain language, together with a statement of uncertainty and a discussion of the limitations of the study, in particular the errors described next.

3.2 Errors in a study

The difference between the conclusions of a study and the truth about the target population comes from three sources.

Definition 3.1 (Study, sample and measurement error)**.**

- **Study error**: the difference between the attribute in the *study population* and the attribute in the *target population*. It cannot be reduced by taking a bigger sample and cannot be quantified from the data; it must be argued from context.
- **Sample error**: the difference between the attribute in the *sample* and the attribute in the *study population*. With a random sampling protocol, sample error can be modelled with probability and quantified (confidence intervals); it decreases as n grows.
- **Measurement error**: the difference between the measured value of a variate and its true value. It may be random (imprecision) or systematic (bias, e.g. a mis-calibrated scale, a leading survey question).

Example 3.2 (Identifying the errors)**.** A researcher wants the average weekly hours of exercise of Ontario university students. She emails a survey to 5000 students chosen at random from the UW registrar's list; 900 reply. Students report their own exercise hours for "a typical week".

- Target population: Ontario university students. Study population: UW students on the registrar's list. *Study error*: UW students may exercise more or less than students elsewhere.
- Sample error: even a perfect random sample of 900 UW students would not have exactly the UW mean. Worse, the 18% who replied are self-selected (*non-response bias*); keen exercisers may be more likely to answer.
- Measurement error: self-reported exercise is typically overstated (systematic) and "typical week" is vague (random).

3.3 Sampling protocols

Definition 3.3 (Simple random sampling)**.** A *simple random sample* (SRS) of size n from a study population of N units is chosen so that every subset of n units has the same probability $1/\binom{N}{n}$ of being selected. It requires a list of the population (a *sampling frame*) and a random number generator.

Random sampling is what justifies the probability models of the later chapters: under an SRS, the sample is "representative on average" and the sample error has a known distribution. Other protocols:

- **Stratified sampling**: divide the population into groups (strata, e.g. faculties) and take an SRS within each. Ensures every stratum is represented and often improves precision.
- **Cluster sampling**: randomly select whole groups (e.g. classrooms) and measure all units in them. Cheaper, but usually less precise.
- **Systematic sampling**: every kth unit from a list after a random start.
- **Convenience / volunteer sampling**: whoever is easiest to reach or chooses to respond. Cheap but prone to serious, unquantifiable bias.

Key distinction

Random *sampling* (selecting units at random) supports generalizing from the sample to the study population. Random *assignment* of treatments in an experiment (Chapter 8) supports causal conclusions. They are different things, and a study may have one, both or neither.

3.4 A complete PPDAC example

Example 3.4 (Fuel economy)**. Problem.** Does a new fuel additive improve the fuel economy (km/L) of a company's delivery vans? Target population: the company's current and future vans in normal use. Response: fuel economy; explanatory variate: additive (yes/no). Causative question.

Plan. Study population: the 40 vans at the Kitchener depot. Randomly assign 20 vans to use the additive for a month, the other 20 not (a better design: use each van in both conditions in random order, giving *paired* data, Chapter 6). Measure fuel economy from fuel receipts and odometer readings, recorded by the same clerk.

Data. Two vans were out of service for a week; their data were based on three weeks. One receipt was lost.

Analysis. Side-by-side boxplots, a two-sample t confidence interval for the difference in mean fuel economy (Chapter 6).

Conclusion. Report the estimated improvement with a confidence interval. Limitations: study error (Kitchener vans, one month in one season may not represent all vans year-round), measurement error in the receipts, small sample.

Common pitfalls

- Increasing the sample size reduces sample error only; it does nothing about study error or systematic measurement error. A huge biased sample is still biased (the 1936 *Literary Digest* poll had over two million responses and predicted the wrong election winner).
- Be precise about the difference between the target and the study population.
- Non-response and volunteer samples break the randomness assumptions behind all later inference.

Summary

- PPDAC: Problem (population, variates, attributes, question), Plan (study population, sampling protocol, measurement), Data, Analysis, Conclusion.
- Study error (target vs study population), sample error (sample vs study population; quantifiable under random sampling), measurement error (measured vs true value).
- Simple random sampling is the basic protocol justifying probability models; stratified, cluster and systematic sampling are variations; convenience samples are unreliable.

Chapter 4

Estimation

4.1 Estimators and sampling distributions

Definition 4.1 (Estimate vs estimator)**.** An *estimate* $\hat{\theta} = g(y_1, \ldots, y_n)$ is a number computed from the observed data. The corresponding *estimator* $\tilde{\theta} = g(Y_1, \ldots, Y_n)$ is a random variable: the rule applied to data that *could* be observed. The distribution of $\tilde{\theta}$ is its *sampling distribution*.

The sampling distribution tells us how much estimates vary from sample to sample, and so how much we can trust one estimate. Two desirable properties: $\tilde{\theta}$ is *unbiased* if $\mathrm{E}(\tilde{\theta}) = \theta$, and its variance should be small.

Theorem 4.2 (Sampling distribution of the sample mean)**.** *If $Y_1, \ldots, Y_n$ are independent with mean μ and standard deviation σ, then $\mathrm{E}(\bar{Y}) = \mu$ and $\mathrm{sd}(\bar{Y}) = \sigma/\sqrt{n}$. If the Y_i are $\mathrm{G}(\mu, \sigma)$, then exactly $\bar{Y} \sim \mathrm{G}(\mu, \sigma/\sqrt{n})$; otherwise, by the Central Limit Theorem, approximately so for large n.*

Theorem 4.3. *$S^2 = \frac{1}{n-1}\sum(Y_i - \bar{Y})^2$ is unbiased for σ^2.*

Proof. $\sum(Y_i - \bar{Y})^2 = \sum Y_i^2 - n\bar{Y}^2$. Using $\mathrm{E}(Y_i^2) = \sigma^2 + \mu^2$ and $\mathrm{E}(\bar{Y}^2) = \sigma^2/n + \mu^2$, $\mathrm{E}\sum(Y_i - \bar{Y})^2 = n(\sigma^2 + \mu^2) - (\sigma^2 + n\mu^2) = (n-1)\sigma^2$. □

4.2 Interval estimation and coverage

A point estimate alone says nothing about precision. An *interval estimate* $[L(y), U(y)]$ gives a range of plausible values.

Definition 4.4 (Coverage probability, confidence interval)**.** For an interval estimator $[L(\mathbf{Y}), U(\mathbf{Y})]$, the *coverage probability* is $C(\theta) = \mathrm{P}\left(L(\mathbf{Y}) \le \theta \le U(\mathbf{Y})\right)$. If $C(\theta) = p$ for all θ (exactly or approximately), the observed interval $[L(y), U(y)]$ is a $100p\%$ *confidence interval* for θ.

Interpreting a confidence interval

A 95% CI does *not* mean "there is a 95% probability that θ is in $[L(y), U(y)]$". Once computed, the interval either contains θ or it does not. The 95% refers to the procedure: if we repeated the study many times and built an interval each time, about 95% of those intervals would contain the true θ.

4.3 Pivotal quantities

Definition 4.5 (Pivotal quantity)**.** A *pivotal quantity* $Q = Q(\mathbf{Y}; \theta)$ is a function of the data and θ whose distribution is completely known (does not depend on θ or other unknown parameters). It is an *asymptotic pivotal quantity* if this holds approximately for large n.

Recipe. Find a, b with $\mathrm{P}(a \le Q \le b) = p$, then rearrange the inequality $a \le Q(\mathbf{Y}; \theta) \le b$ into $L(\mathbf{Y}) \le \theta \le U(\mathbf{Y})$ and plug in the data.

Example 4.6 (Gaussian mean, σ known)**.** If $Y_i \sim \mathrm{G}(\mu, \sigma)$ with σ known, $Z = \frac{\bar{Y}-\mu}{\sigma/\sqrt{n}} \sim \mathrm{G}(0,1)$ is pivotal. From $\mathrm{P}(-1.96 \le Z \le 1.96) = 0.95$:

$$\bar{y} \pm 1.96 \frac{\sigma}{\sqrt{n}} \quad \text{is a 95\% CI for } \mu.$$

For the fill-weight data of Chapter 2 ($n = 10$, $\bar{y} = 12.52$), if $\sigma = 0.6$ were known, the CI is $12.52 \pm 1.96(0.6/\sqrt{10}) = 12.52 \pm 0.372 = [12.148,\ 12.892]$.

General form: *estimate* $\pm$ (table value) $\times$ (standard error). Useful $\mathrm{G}(0,1)$ values: $\mathrm{P}(|Z| \le 1.645) = 0.90$, $\mathrm{P}(|Z| \le 1.960) = 0.95$, $\mathrm{P}(|Z| \le 2.576) = 0.99$.

4.4 Approximate confidence intervals via the CLT

Theorem 4.7 (Approximate CIs for binomial and Poisson)**.** *1. If* $Y \sim \mathrm{Binomial}(n, \theta)$*, then*

$$\frac{\tilde{\theta} - \theta}{\sqrt{\theta(1-\theta)/n}}$$

is approximately $\mathrm{G}(0,1)$*, and so is the same quantity with* θ *replaced by* $\tilde{\theta}$ *in the denominator. An approximate* 95% *CI is*

$$\hat{\theta} \pm 1.96 \sqrt{\hat{\theta}(1-\hat{\theta})/n}.$$

2. If $Y_1, \ldots, Y_n$ *are independent* $\mathrm{Poisson}(\theta)$*, an approximate* 95% *CI is* $\hat{\theta} \pm 1.96\sqrt{\hat{\theta}/n}$ *with* $\hat{\theta} = \bar{y}$*.*

Example 4.8.

- Survey of Chapter 2 ($n = 200$, $\hat{\theta} = 0.29$): $\sqrt{0.29(0.71)/200} = 0.03209$, so the CI is $0.29 \pm 0.0629 = [0.227,\ 0.353]$.
- Help-desk calls ($n = 10$, $\hat{\theta} = 3.5$): $\sqrt{3.5/10} = 0.5916$, so the CI is $3.5 \pm 1.160 = [2.340,\ 4.660]$. (With $n = 10$, the approximation is rough.)

4.5 Likelihood intervals

Definition 4.9 (Likelihood interval)**.** A $100p\%$ *likelihood interval* for θ is $\{\theta : R(\theta) \ge p\}$. Values with $R(\theta) \ge 0.5$ are very plausible, $0.1 \le R < 0.5$ plausible, $0.01 \le R < 0.1$ implausible, and $R < 0.01$ very implausible.

Theorem 4.10 (Likelihood ratio statistic)**.** *Let* $\Lambda(\theta) = -2\log\left(L(\theta)/L(\tilde{\theta})\right)$ *(with* $\tilde{\theta}$ *the ML estimator). Under regularity conditions, for large* n*,* $\Lambda(\theta)$ *is approximately* $\chi^2(1)$ *when* θ *is the true value. Hence it is an asymptotic pivotal quantity, and*

$$\mathrm{P}\left(R(\theta) \ge p\right) = \mathrm{P}\left(\Lambda(\theta) \le -2\log p\right) \approx \mathrm{P}\left(W \le -2\log p\right), \quad W \sim \chi^2(1).$$

Consequences (using $\mathrm{P}(W \le w) = \mathrm{P}(|Z| \le \sqrt{w}) = 2\Phi(\sqrt{w}) - 1$):

likelihood interval	$-2\log p$	approximate confidence
50%	1.386	76%
15%	3.794	95%
10%	4.605	96.8%
1%	9.210	99.8%

Exactly, a 95% CI corresponds to $p = e^{-3.841/2} = 0.1465$, so a **15% likelihood interval is an approximate 95% confidence interval**.

Example 4.11 (Comparing intervals)**.** Solving $R(\theta) = 0.15$ numerically:

- survey: 15% LI $\approx [0.230,\ 0.355]$ vs CLT interval $[0.227,\ 0.353]$;
- help-desk calls: 15% LI $\approx [2.466,\ 4.790]$ vs CLT interval $[2.340,\ 4.660]$.

The likelihood interval is not symmetric about $\hat\theta$: it follows the skewness of the likelihood, and is generally more accurate for small n and parameters near a boundary (e.g. θ near 0).

4.6 The chi-squared and t distributions

Definition 4.12 (Chi-squared)**.** $W \sim \chi^2(k)$ (k *degrees of freedom*) has p.d.f.

$$f(w) = \frac{1}{2^{k/2}\Gamma(k/2)} w^{k/2-1} e^{-w/2}, \qquad w > 0.$$

Then $\mathrm{E}(W) = k$, $\mathrm{Var}(W) = 2k$.

Theorem 4.13. *1. If $Z \sim \mathrm{G}(0,1)$ then $Z^2 \sim \chi^2(1)$.*

2. If $W_1, \dots, W_m$ are independent with $W_i \sim \chi^2(k_i)$, then $\sum W_i \sim \chi^2(\sum k_i)$. In particular, if $Z_1, \dots, Z_k$ are independent $\mathrm{G}(0,1)$, $\sum Z_i^2 \sim \chi^2(k)$.

3. $\chi^2(2)$ is the exponential distribution with mean 2. Hence if $Y_i \sim \mathrm{Exponential}(\theta)$ independent, $2\sum Y_i/\theta \sim \chi^2(2n)$.

Definition 4.14 (Student t)**.** If $Z \sim \mathrm{G}(0,1)$ and $W \sim \chi^2(k)$ are independent, then $T = \frac{Z}{\sqrt{W/k}} \sim t(k)$. The $t(k)$ density is symmetric about 0, bell shaped, with heavier tails than $\mathrm{G}(0,1)$; as $k \to \infty$, $t(k) \to \mathrm{G}(0,1)$.

Theorem 4.15 (Gaussian sampling theory)**.** *If $Y_1, \dots, Y_n$ are independent $\mathrm{G}(\mu, \sigma)$, then $\bar Y$ and S^2 are independent,*

$$\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2}\sum (Y_i - \bar Y)^2 \sim \chi^2(n-1), \qquad T = \frac{\bar Y - \mu}{S/\sqrt{n}} \sim t(n-1).$$

Proof sketch. The independence and the $\chi^2(n-1)$ result are proved in Stat 330 (one degree of freedom is "used up" estimating μ by $\bar Y$). Then $T = \frac{(\bar Y - \mu)/(\sigma/\sqrt{n})}{\sqrt{[(n-1)S^2/\sigma^2]/(n-1)}}$ is $\mathrm{G}(0,1)$ over $\sqrt{\chi^2(n-1)/(n-1)}$ with independent numerator and denominator. □

4.7 Confidence intervals for Gaussian models

Theorem 4.16. *For a* $\mathrm{G}(\mu, \sigma)$ *sample with both parameters unknown:*

- *a* $100p\%$ *CI for* μ *is* $\bar{y} \pm a\, s/\sqrt{n}$ *where* $\mathrm{P}(T \le a) = (1+p)/2$, $T \sim t(n-1)$;
- *a* $100p\%$ *CI for* σ^2 *is* $\left[\dfrac{(n-1)s^2}{b},\ \dfrac{(n-1)s^2}{a}\right]$ *where* $\mathrm{P}(W \le a) = (1-p)/2$ *and* $\mathrm{P}(W \le b) = (1+p)/2$, $W \sim \chi^2(n-1)$; *take square roots for a CI for* σ.

Example 4.17 (Fill weights)**.** $n = 10$, $\bar{y} = 12.52$, $(n-1)s^2 = S_{yy} = 3.456$, $s^2 = 0.384$, $s = 0.6197$.

Mean. From t tables, $\mathrm{P}(T \le 2.262) = 0.975$ for $T \sim t(9)$. Standard error $s/\sqrt{10} = 0.19596$, so the 95% CI is

$$12.52 \pm 2.262(0.19596) = 12.52 \pm 0.443 = [12.077,\ 12.963].$$

Variance. For $W \sim \chi^2(9)$: $\mathrm{P}(W \le 2.700) = 0.025$, $\mathrm{P}(W \le 19.023) = 0.975$. The 95% CI for σ^2 is

$$\left[\frac{3.456}{19.023},\ \frac{3.456}{2.700}\right] = [0.1817,\ 1.280],$$

and for σ: $[0.426,\ 1.131]$. Note how wide it is: 10 observations say little about variability. The interval is not symmetric about s^2.

Example 4.18 (Exact CI for an exponential mean)**.** For the 8 component lifetimes of Chapter 2, $\sum y_i = 24$. Since $2\sum Y_i/\theta \sim \chi^2(16)$ with $\mathrm{P}(W \le 6.908) = 0.025$ and $\mathrm{P}(W \le 28.845) = 0.975$, from $6.908 \le 48/\theta \le 28.845$ we get the 95% CI

$$\left[\frac{48}{28.845},\ \frac{48}{6.908}\right] = [1.664,\ 6.949] \text{ years}.$$

4.8 Choosing the sample size

To estimate a parameter with a 95% CI of half-width (margin of error) at most ℓ:

Binomial proportion. Need $1.96\sqrt{\theta(1-\theta)/n} \le \ell$, i.e. $n \ge (1.96/\ell)^2\,\theta(1-\theta)$. Since $\theta(1-\theta) \le 1/4$, the conservative choice is $n \ge (1.96/\ell)^2/4$.

Example 4.19. For a margin of $\ell = 0.03$: $(1.96/0.03)^2 = 4268.4$, so the conservative n is $4268.4/4 = 1067.1$, i.e. $n = 1068$ (this is why national polls of about 1000 people are quoted as "accurate to within 3 percentage points, 19 times out of 20"). If we believe $\theta \approx 0.29$, then $n \ge 4268.4(0.29)(0.71) = 878.9$, so $n = 879$ suffices.

Gaussian mean. With a guess for σ, $n \ge (1.96\sigma/\ell)^2$. For $\sigma \approx 0.6$ and $\ell = 0.2$, $n \ge (5.88)^2 = 34.57$, so $n = 35$. (Strictly, s and t replace σ and 1.96, but for planning this is fine.)

Common pitfalls

- Always round sample sizes *up*.
- Use $t(n-1)$, not $\mathrm{G}(0,1)$, when σ is estimated by s in a Gaussian model (matters for small n).
- The chi-squared CI for σ^2 uses *both* tails with different table values; it is not "estimate $\pm$ something".
- Confidence statements quantify sample error only, not study or measurement error.

Summary

- An estimator is random and has a sampling distribution; $\mathrm{sd}(\bar{Y}) = \sigma/\sqrt{n}$.
- CIs come from (asymptotic) pivotal quantities; the confidence level is the long-run coverage of the procedure.
- $-2\log R(\theta) \approx \chi^2(1)$: a 15% likelihood interval is an approximate 95% CI.
- Gaussian: $\frac{\bar{Y}-\mu}{S/\sqrt{n}} \sim t(n-1)$, $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$. Exponential: $2\sum Y_i/\theta \sim \chi^2(2n)$.
- Sample size for proportion: $n \geq (1.96/\ell)^2\theta(1-\theta)$, at most $(1.96/\ell)^2/4$.

Chapter 5

Tests of Hypotheses

5.1 The logic of a test

A *test of hypothesis* asks whether the data are consistent with a specific claim about the model. The claim being tested is the *null hypothesis* H_0, e.g. $H_0 : \theta = \theta_0$. The approach is a kind of proof by contradiction: assume H_0 is true and ask how surprising the observed data would be. If very surprising, we have evidence against H_0.

Definition 5.1 (Test statistic / discrepancy measure)**.** A *test statistic* (or *discrepancy measure*) $D = D(\mathbf{Y})$ is a function of the data, constructed so that $D \geq 0$, $D = 0$ (or small) indicates the best agreement with H_0, and large values indicate disagreement. Its distribution under H_0 must be known (exactly or approximately). The observed value is $d = D(y)$.

Definition 5.2 (p-value)**.** The *p-value* (observed significance level) is

$$p = \mathrm{P}(D \geq d;\ H_0),$$

the probability, computed assuming H_0 is true, of observing a discrepancy at least as large as the one actually observed.

Interpreting p-values

p-value	interpretation
$p > 0.10$	no evidence against H_0
$0.05 < p \leq 0.10$	weak evidence against H_0
$0.01 < p \leq 0.05$	evidence against H_0
$0.001 < p \leq 0.01$	strong evidence against H_0
$p \leq 0.001$	very strong evidence against H_0

A small p-value means the data are unlikely if H_0 is true. A large p-value does *not* prove H_0; it means the data are consistent with it (we never "accept" H_0). The p-value is not the probability that H_0 is true.

Remark. Many texts phrase tests as decisions: reject H_0 at level α if $p \leq \alpha$. A *type I error* is rejecting a true H_0 (probability α); a *type II error* is failing to reject a false H_0. Stat 231 emphasizes reporting the p-value as a measure of evidence, together with a confidence interval for the size of the effect. Statistical significance is not practical significance: with a huge n a tiny, unimportant effect can produce a tiny p-value.

5.2 Tests for binomial and Poisson parameters

Binomial. For $Y \sim \text{Binomial}(n, \theta)$ and $H_0 : \theta = \theta_0$, use $D = |Y - n\theta_0|$. Exact p-value: $\text{P}(|Y - n\theta_0| \geq d)$ summing binomial probabilities. Normal approximation: $p \approx \text{P}\left(|Z| \geq d/\sqrt{n\theta_0(1-\theta_0)}\right)$.

Example 5.3 (Binomial test)**.** In the survey of Chapter 2, $y = 58$ of $n = 200$ students work part time. A report claims 25% do. Test $H_0 : \theta = 0.25$. Under H_0, $\text{E}(Y) = 50$ and $\text{sd}(Y) = \sqrt{200(0.25)(0.75)} = \sqrt{37.5} = 6.124$. Observed $d = |58 - 50| = 8$, so

$$p \approx \text{P}(|Z| \geq 8/6.124) = \text{P}(|Z| \geq 1.306) = 2(1 - 0.9043) = 0.19.$$

(The exact binomial sum gives $p = 0.220$.) No evidence against H_0: the data are consistent with $\theta = 0.25$.

Poisson. For $Y_1, \ldots, Y_n$ independent Poisson(θ) and $H_0 : \theta = \theta_0$, $\sum Y_i \sim \text{Poisson}(n\theta_0)$ under H_0; use $D = |\sum Y_i - n\theta_0|$ with either exact Poisson probabilities or the approximation $\sum Y_i \approx \text{G}(n\theta_0, \sqrt{n\theta_0})$.

Example 5.4 (Poisson test)**.** The help desk was designed for $\theta_0 = 2.5$ calls per period. Observed $\sum y_i = 35$ in $n = 10$ periods; under H_0, $\sum Y_i \sim \text{Poisson}(25)$. Then $d = |35 - 25| = 10$ and $p \approx \text{P}(|Z| \geq 10/5) = \text{P}(|Z| \geq 2) = 0.0455$: evidence against H_0; the call rate seems higher than designed for. (The likelihood ratio test below gives $p \approx 0.059$, weak evidence: with only $n = 10$ the approximations differ, so the conclusion is "borderline" evidence.)

5.3 Tests for Gaussian parameters

For a $\text{G}(\mu, \sigma)$ sample:

- $H_0 : \mu = \mu_0$: use $D = \dfrac{|\bar{Y} - \mu_0|}{S/\sqrt{n}}$; under H_0, $\frac{\bar{Y} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$, so $p = 2\big(1 - \text{P}(T \leq d)\big)$.
- $H_0 : \sigma = \sigma_0$: use $U = (n-1)S^2/\sigma_0^2 \sim \chi^2(n-1)$ under H_0. For observed u, the two-sided p-value is $p = 2\min\big\{\text{P}(U \leq u),\ \text{P}(U \geq u)\big\}$.

Example 5.5 (t test for a mean)**.** Fill weights: $n = 10$, $\bar{y} = 12.52$, $s = 0.6197$. The label says 12 g on average; test $H_0 : \mu = 12$.

$$d = \frac{|12.52 - 12|}{0.6197/\sqrt{10}} = \frac{0.52}{0.19596} = 2.654.$$

From $t(9)$ tables, $\text{P}(T \leq 2.262) = 0.975$ and $\text{P}(T \leq 2.821) = 0.99$, so $0.01 < \text{P}(T \geq 2.654) < 0.025$ and $0.02 < p < 0.05$ (software: $p = 0.026$). There is evidence against H_0: the mean fill weight appears to exceed 12 g.

Example 5.6 (Test for a standard deviation)**.** The machine is supposed to have $\sigma = 1$. Here $u = (n-1)s^2/1^2 = 3.456$. From $\chi^2(9)$ tables, $\text{P}(U \leq 3.325) = 0.05$ and $\text{P}(U \leq 4.168) = 0.10$, so $0.05 < \text{P}(U \leq 3.456) < 0.10$ and $0.10 < p < 0.20$ (software: $\text{P}(U \leq 3.456) = 0.0566$, $p = 0.113$). No evidence against $\sigma = 1$, despite $s = 0.62$, because n is small.

5.4 The likelihood ratio test

Definition 5.7 (Likelihood ratio test statistic)**.** For $H_0 : \theta = \theta_0$ with a single parameter,

$$\Lambda = -2\log\frac{L(\theta_0)}{L(\hat\theta)} = -2\log R(\theta_0).$$

Large Λ (small $R(\theta_0)$) is evidence against H_0. Under H_0, $\Lambda \approx \chi^2(1)$ for large n, so with observed λ,

$$p \approx \mathrm{P}(W \ge \lambda) = 2\big(1 - \Phi(\sqrt{\lambda})\big), \qquad W \sim \chi^2(1).$$

More generally, if H_0 fixes k of the parameters (reducing the dimension of the parameter space by k), $\Lambda = -2\log\big(\max_{H_0} L / \max L\big) \approx \chi^2(k)$.

Example 5.8 (LRT for the binomial and Poisson examples)**. Binomial**, $H_0 : \theta = 0.25$, $\hat\theta = 0.29$:

$$\lambda = 2\left[58\log\frac{0.29}{0.25} + 142\log\frac{0.71}{0.75}\right] = 2[8.6085 - 7.7827] = 1.651,$$

$p \approx \mathrm{P}(|Z| \ge \sqrt{1.651}) = \mathrm{P}(|Z| \ge 1.285) = 0.199$, matching the earlier conclusion.

Poisson, $H_0 : \theta = 2.5$, $\hat\theta = 3.5$, $\ell(\theta) = 35\log\theta - 10\theta$:

$$\lambda = 2\big[35\log(3.5/2.5) + 10(2.5 - 3.5)\big] = 2[11.777 - 10] = 3.553,$$

$p \approx \mathrm{P}(|Z| \ge 1.885) = 0.059$.

Example 5.9 (LRT for a Gaussian mean)**.** For $H_0 : \mu = \mu_0$ with σ unknown, the maximized likelihoods give $\Lambda = n\log(\hat\sigma_0^2/\hat\sigma^2)$, where $\hat\sigma_0^2 = \frac{1}{n}\sum(y_i - \mu_0)^2 = \hat\sigma^2 + (\bar y - \mu_0)^2$. Thus $\Lambda = n\log\big(1 + \frac{t^2}{n-1}\big)$ where t is the t statistic: the LRT is equivalent to the t test. For the fill weights: $\lambda = 10\log(1 + 2.654^2/9) = 10\log(1.7824) = 5.78$, $p \approx 0.016$. The exact t test ($p = 0.026$) is preferred for small n.

5.5 Tests and confidence intervals

Theorem 5.10 (Duality)**.** *Suppose $[L, U]$ is a $100p\%$ CI built from a pivotal quantity $Q(\mathbf{Y};\theta)$ and the test of $H_0 : \theta = \theta_0$ uses the discrepancy based on the same $Q(\mathbf{Y};\theta_0)$. Then the p-value is at least $1 - p$ if and only if θ_0 lies in the $100p\%$ CI. In particular, $H_0 : \theta = \theta_0$ has p-value ≥ 0.05 exactly when θ_0 is inside the 95% CI.*

Proof sketch for the t case. $\mu_0 \in \bar y \pm as/\sqrt{n} \iff |\bar y - \mu_0|/(s/\sqrt{n}) \le a \iff d \le a \iff \mathrm{P}(|T| \ge d) \ge \mathrm{P}(|T| \ge a) = 1 - p.$ □

Example 5.11. The 95% CI for the mean fill weight was $[12.077, 12.963]$, which excludes 12: consistent with $p = 0.026 < 0.05$. The approximate 95% CI $[0.227, 0.353]$ for the binomial proportion contains 0.25: consistent with $p \approx 0.19$. Equivalently, the set of θ_0 whose LRT p-value is at least 0.05 is (approximately) the 15% likelihood interval.

Common pitfalls

- Never say "we accept H_0" or "H_0 is true"; say the data provide no evidence against it.
- The p-value is computed *assuming* H_0; it is not $\mathrm{P}(H_0 \mid \text{data})$.
- Formulate H_0 before looking at the data; testing many hypotheses and reporting the smallest p-value is misleading.
- Report a CI alongside the p-value to show the size of the effect.

Summary

- Test statistic D, observed d, p-value $= \mathrm{P}(D \geq d; H_0)$; interpret on the evidence scale.
- Binomial/Poisson: $D = |Y - \mathrm{E}_0(Y)|$ with exact or normal-approximation p-values. Gaussian mean: $t(n-1)$; variance: $\chi^2(n-1)$.
- LRT: $\Lambda = -2 \log R(\theta_0) \approx \chi^2(1)$ (or $\chi^2(k)$).
- $p \geq 0.05$ iff θ_0 is in the corresponding 95% CI.

Chapter 6

Gaussian Response Models

6.1 Introduction

A *Gaussian response model* has independent responses

$$Y_i \sim \mathrm{G}\left(\mu(x_i),\ \sigma\right), \qquad i = 1, \ldots, n,$$

where the mean $\mu(x_i)$ depends on explanatory variate(s) x_i through unknown parameters and the standard deviation σ is the same for all units. Equivalently $Y_i = \mu(x_i) + R_i$ with $R_i \sim \mathrm{G}(0, \sigma)$ independent errors. Important special cases:

- $\mu(x_i) = \mu$: the single Gaussian sample of Chapters 4–5;
- $\mu(x_i) = \alpha + \beta x_i$: **simple linear regression**;
- two groups with means μ_1, μ_2: the **two-sample** problem.

6.2 Simple linear regression

Definition 6.1 (Model)**.** $Y_i \sim \mathrm{G}(\alpha+\beta x_i, \sigma)$ independently, with the x_i treated as known constants. β is the change in mean response per unit increase in x; α is the mean response at $x = 0$.

Notation: $S_{xx} = \sum(x_i - \bar{x})^2$, $S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$, $S_{yy} = \sum(y_i - \bar{y})^2$.

Theorem 6.2 (Least squares = maximum likelihood)**.** *The MLEs of α and β minimize the sum of squared deviations $\sum(y_i - \alpha - \beta x_i)^2$ and are*

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}.$$

The MLE of σ^2 is $\frac{1}{n}\sum \hat{r}_i^2$ with residuals $\hat{r}_i = y_i - \hat{\alpha} - \hat{\beta}x_i$; we use instead the unbiased estimate

$$s_e^2 = \frac{1}{n-2}\sum_{i=1}^{n} \hat{r}_i^2 = \frac{S_{yy} - \hat{\beta} S_{xy}}{n-2}.$$

Proof. $\ell(\alpha, \beta, \sigma) = -n \log \sigma - \frac{1}{2\sigma^2}\sum(y_i - \alpha - \beta x_i)^2$, so for every σ, maximizing over (α, β) is the same as minimizing $Q = \sum(y_i - \alpha - \beta x_i)^2$. Setting $\partial Q/\partial\alpha = -2\sum(y_i - \alpha - \beta x_i) = 0$ gives $\alpha = \bar{y} - \beta\bar{x}$. Substituting and setting $\partial Q/\partial\beta = -2\sum x_i(y_i - \alpha - \beta x_i) = 0$ gives $\sum x_i\big[(y_i - \bar{y}) - \beta(x_i - \bar{x})\big] = 0$, i.e. $S_{xy} = \beta S_{xx}$ (since $\sum \bar{x}(y_i - \bar{y}) = 0$ etc.). For the residual sum of squares, $\sum \hat{r}_i^2 = \sum[(y_i - \bar{y}) - \hat{\beta}(x_i - \bar{x})]^2 = S_{yy} - 2\hat{\beta}S_{xy} + \hat{\beta}^2 S_{xx} = S_{yy} - \hat{\beta}S_{xy}$. □

The divisor $n-2$ reflects the two estimated mean parameters, just as $n-1$ does for a single sample.

6.3 Inference for the slope, intercept and mean response

Theorem 6.3 (Sampling distributions)**.** *Writing $\tilde\beta = \sum (x_i-\bar x)Y_i/S_{xx}$, a linear combination of independent Gaussians,*

$$\tilde\beta \sim \mathrm{G}\left(\beta,\ \frac{\sigma}{\sqrt{S_{xx}}}\right), \qquad \tilde\alpha \sim \mathrm{G}\left(\alpha,\ \sigma\sqrt{\tfrac{1}{n}+\tfrac{\bar x^2}{S_{xx}}}\right), \qquad \frac{(n-2)S_e^2}{\sigma^2} \sim \chi^2(n-2),$$

with S_e^2 independent of $(\tilde\alpha,\tilde\beta)$. Hence

$$\frac{\tilde\beta-\beta}{S_e/\sqrt{S_{xx}}} \sim t(n-2).$$

More generally, the estimator $\tilde\mu(x)=\tilde\alpha+\tilde\beta x$ of the mean response $\mu(x)=\alpha+\beta x$ satisfies

$$\frac{\tilde\mu(x)-\mu(x)}{S_e\sqrt{\frac{1}{n}+\frac{(x-\bar x)^2}{S_{xx}}}} \sim t(n-2).$$

Consequently, with a chosen so that $\mathrm{P}(T\le a)=(1+p)/2$ for $T\sim t(n-2)$:

- CI for β: $\hat\beta \pm a\, s_e/\sqrt{S_{xx}}$; test $H_0:\beta=\beta_0$ with $d=|\hat\beta-\beta_0|/(s_e/\sqrt{S_{xx}})$. The test of $\beta=0$ asks whether there is any (linear) relationship at all.
- CI for α: $\hat\alpha \pm a\, s_e\sqrt{1/n+\bar x^2/S_{xx}}$.
- CI for the mean response at x: $\hat\mu(x) \pm a\, s_e\sqrt{\frac{1}{n}+\frac{(x-\bar x)^2}{S_{xx}}}$.

Theorem 6.4 (Prediction interval)**.** *A new response at x is $Y=\mu(x)+R$ with R independent of the data. Then $Y-\tilde\mu(x)\sim \mathrm{G}\left(0,\sigma\sqrt{1+\frac{1}{n}+\frac{(x-\bar x)^2}{S_{xx}}}\right)$, and a* $100p\%$ prediction interval *for Y is*

$$\hat\mu(x) \pm a\, s_e\sqrt{1+\frac{1}{n}+\frac{(x-\bar x)^2}{S_{xx}}}.$$

The extra "1+" accounts for the variability of the individual new unit about its mean; a prediction interval is always wider than the CI for the mean response and does not shrink to zero width as $n\to\infty$. Both intervals are narrowest at $x=\bar x$. Predicting far outside the range of observed x (*extrapolation*) is risky because the linear model may not hold there.

Example 6.5 (A complete regression analysis)**.** A chemical process was run at $n=10$ temperature settings x (in units of 10°C) and the yield y (grams) recorded:

x	2	3	5	6	8	9	11	12	14	15
y	5.1	6.0	8.3	8.9	11.2	12.0	14.6	15.1	17.8	18.4

Summaries. $\sum x=85$, $\sum y=117.4$, $\sum x^2=905$, $\sum y^2=1576.12$, $\sum xy=1187.7$, so $\bar x=8.5$, $\bar y=11.74$ and

$$S_{xx}=905-\tfrac{85^2}{10}=182.5, \quad S_{xy}=1187.7-\tfrac{85(117.4)}{10}=189.8, \quad S_{yy}=1576.12-\tfrac{117.4^2}{10}=197.844.$$

Estimates. $\hat\beta = 189.8/182.5 = 1.04$, $\hat\alpha = 11.74 - 1.04(8.5) = 2.90$, so the fitted line is $\hat y = 2.90 + 1.04x$. Residual sum of squares $197.844 - 1.04(189.8) = 0.452$, so $s_e^2 = 0.452/8 = 0.0565$ and $s_e = 0.2377$.

Slope. $\mathrm{P}(T \le 2.306) = 0.975$ for $T \sim t(8)$. Standard error $s_e/\sqrt{S_{xx}} = 0.2377/13.509 = 0.01760$. 95% CI: $1.04 \pm 2.306(0.01760) = 1.04 \pm 0.0406 = [0.999,\ 1.081]$. For $H_0 : \beta = 0$, $d = 1.04/0.0176 = 59.1$: p is essentially 0, overwhelming evidence of a relationship. For $H_0 : \beta = 1$, $d = 0.04/0.0176 = 2.27$; since $\mathrm{P}(T \le 1.860) = 0.95$ and $\mathrm{P}(T \le 2.306) = 0.975$, $0.05 < p < 0.10$ (software: 0.053), weak evidence against $\beta = 1$, consistent with 1 being just inside the 95% CI.

Intercept. Standard error $0.2377\sqrt{0.1 + 72.25/182.5} = 0.1674$; CI $2.90 \pm 2.306(0.1674) = [2.514,\ 3.286]$.

Mean response at $x = 10$. $\hat\mu(10) = 2.90 + 10.4 = 13.30$, standard error

$$0.2377\sqrt{0.1 + 2.25/182.5} = 0.2377(0.3352) = 0.0797;$$

95% CI $13.30 \pm 0.184 = [13.12,\ 13.48]$.

Prediction at $x = 10$. Standard error $0.2377\sqrt{1.1123} = 0.2507$; 95% PI $13.30 \pm 0.578 = [12.72,\ 13.88]$, much wider than the CI for the mean.

6.4 Checking the model: residual plots

The model assumes (1) linearity of the mean in x, (2) constant σ, (3) Gaussian errors, (4) independence. Check using the residuals $\hat r_i$ or the *standardized residuals* $\hat r_i/s_e$:

- **Residuals vs x_i (or vs fitted values $\hat\mu_i$):** should look like a horizontal band of random scatter around 0. A curved pattern indicates non-linearity; a funnel shape indicates non-constant variance.
- **QQ plot of residuals:** roughly a straight line if errors are Gaussian.
- **Residuals vs time order:** trends or runs suggest dependence.
- About 95% of standardized residuals should lie in $[-2, 2]$; values beyond ± 3 are possible outliers.

Example 6.6. In Example 6.5 the residuals are

$$0.12,\ -0.02,\ 0.20,\ -0.24,\ -0.02,\ -0.26,\ 0.26,\ -0.28,\ 0.34,\ -0.10.$$

They sum to 0 (as least squares residuals always do), show no curvature or funnel, and all standardized residuals are within ± 1.5. The model looks adequate (though with $n = 10$ only gross violations would be visible).

6.5 Comparing two population means

6.5.1 Two independent samples

Definition 6.7 (Two-sample model)**.** $Y_{1j} \sim \mathrm{G}(\mu_1, \sigma)$, $j = 1, \dots, n_1$ and $Y_{2j} \sim \mathrm{G}(\mu_2, \sigma)$, $j = 1, \dots, n_2$, all independent, with a *common* σ.

Theorem 6.8. *With the* pooled *variance estimate* $s_p^2 = \dfrac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$,

$$\frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1+n_2-2).$$

So a CI for $\mu_1-\mu_2$ *is* $\bar{y}_1 - \bar{y}_2 \pm a\, s_p\sqrt{1/n_1+1/n_2}$, *and* $H_0 : \mu_1 = \mu_2$ *is tested with* $d = |\bar{y}_1 - \bar{y}_2|/(s_p\sqrt{1/n_1+1/n_2})$.

If the variances look very different, use the approximate pivotal quantity with $\sqrt{s_1^2/n_1 + s_2^2/n_2}$ in the denominator (approximately $\mathrm{G}(0,1)$ for large samples).

Example 6.9 (Two fertilizers)**.** Plant heights (cm) after six weeks, 8 plants per fertilizer, randomly assigned:

A	24.1	22.8	25.6	23.9	26.2	24.7	23.3	25.0
B	21.9	23.4	22.1	20.8	23.0	22.6	21.5	22.3

$\bar{y}_1 = 195.6/8 = 24.45$, $\bar{y}_2 = 177.6/8 = 22.20$; $\sum(y_{1j}-\bar{y}_1)^2 = 9.22$, $\sum(y_{2j}-\bar{y}_2)^2 = 4.80$. Pooled: $s_p^2 = (9.22+4.80)/14 = 1.0014$, $s_p = 1.0007$. Standard error $s_p\sqrt{1/8+1/8} = 0.5004$. For $t(14)$, $\mathrm{P}(T \le 2.145) = 0.975$:

$$95\%\text{ CI for } \mu_1-\mu_2 :\ 2.25 \pm 2.145(0.5004) = 2.25 \pm 1.073 = [1.18,\ 3.32].$$

For $H_0 : \mu_1 = \mu_2$, $d = 2.25/0.5004 = 4.50$; since $\mathrm{P}(T \le 4.140) = 0.9995$ for $T \sim t(14)$, $p < 0.001$ (software: $p = 0.0005$). Very strong evidence that fertilizer A gives taller plants on average.

6.5.2 Paired data

When each unit is measured twice (before/after) or units come in natural matched pairs (twins, left/right eye), the two measurements are *not* independent. Analyze the differences $d_i = y_{1i} - y_{2i}$ as a single Gaussian sample $D_i \sim \mathrm{G}(\mu_d, \sigma_d)$:

$$\frac{\bar{D} - \mu_d}{S_d/\sqrt{n}} \sim t(n-1).$$

Example 6.10 (Paired vs unpaired analysis)**.** Systolic blood pressure of 8 patients before and after a drug:

before	142	138	151	147	135	160	144	149
after	136	135	144	145	131	152	141	142
difference	6	3	7	2	4	8	3	7

$\bar{d} = 40/8 = 5$, $\sum(d_i-\bar{d})^2 = 36$, $s_d^2 = 36/7 = 5.143$, $s_d = 2.268$, standard error $2.268/\sqrt{8} = 0.802$. For $t(7)$, $\mathrm{P}(T \le 2.365) = 0.975$: CI $5 \pm 2.365(0.802) = 5 \pm 1.90 = [3.10,\ 6.90]$. For $H_0 : \mu_d = 0$, $d = 5/0.802 = 6.24$; $\mathrm{P}(T \le 5.408) = 0.9995$ so $p < 0.001$ (software 0.0004).

Wrong analysis: treating before and after as independent samples gives means 145.75 and 140.75, $s_p = 7.285$, standard error $7.285\sqrt{2/8} = 3.643$, $d = 5/3.643 = 1.37$ on 14 degrees of freedom, $p = 0.19$: no evidence! The large patient-to-patient variation swamps the effect. Pairing removes this variation; this is why paired designs are so efficient.

Common pitfalls

- Paired data must be analyzed as differences; independent samples must not be. Decide from the *design*, not from which gives the smaller p-value.
- Use $n-2$ degrees of freedom in regression, n_1+n_2-2 for pooled two-sample, $n-1$ for paired.
- A CI for the mean response is not a prediction interval for an individual.
- A significant slope in observational data does not show that changing x causes y to change (Chapter 8).

Summary

- $Y_i \sim \mathrm{G}(\alpha+\beta x_i, \sigma)$: $\hat{\beta} = S_{xy}/S_{xx}$, $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$, $s_e^2 = (S_{yy} - \hat{\beta}S_{xy})/(n-2)$.
- Inference uses $t(n-2)$ with standard errors $s_e/\sqrt{S_{xx}}$ (slope), $s_e\sqrt{1/n + (x-\bar{x})^2/S_{xx}}$ (mean response), and $s_e\sqrt{1 + 1/n + (x-\bar{x})^2/S_{xx}}$ (prediction).
- Check assumptions with residual plots and QQ plots.
- Two independent samples: pooled $t(n_1+n_2-2)$. Paired: one-sample $t(n-1)$ on differences.

Chapter 7

Tests and Inference Problems Based on Multinomial Models

7.1 The multinomial model

Suppose each of n independent units falls into exactly one of k categories, with probabilities $\theta_1, \dots, \theta_k$ ($\sum \theta_j = 1$). The category counts $(Y_1, \dots, Y_k)$, $\sum Y_j = n$, have the *multinomial* distribution

$$\mathrm{P}(Y_1 = y_1, \dots, Y_k = y_k) = \frac{n!}{y_1! \cdots y_k!} \theta_1^{y_1} \cdots \theta_k^{y_k}.$$

Each $Y_j \sim \text{Binomial}(n, \theta_j)$ marginally.

Theorem 7.1 (Multinomial MLE)**.** *The MLEs are $\hat{\theta}_j = y_j / n$.*

Proof. Maximize $\ell(\theta) = \sum y_j \log \theta_j$ subject to $\sum \theta_j = 1$. With a Lagrange multiplier, $y_j / \theta_j = \lambda$ for all j, so $\theta_j \propto y_j$, and the constraint gives $\theta_j = y_j / n$. □

7.2 Likelihood ratio goodness-of-fit test

We want to test whether the data are consistent with a hypothesized model $H_0 : \theta_j = \theta_j(\alpha)$ for $j = 1, \dots, k$, where α is a vector of p unknown parameters (possibly $p = 0$, a completely specified model).

Theorem 7.2 (Likelihood ratio GOF statistic)**.** *Let $\hat{\alpha}$ be the MLE of α under H_0 and $e_j = n\, \theta_j(\hat{\alpha})$ the* expected frequencies*. The likelihood ratio statistic is*

$$\Lambda = -2 \log \frac{L(\theta(\hat{\alpha}))}{L(\hat{\theta})} = 2 \sum_{j=1}^{k} y_j \log \frac{y_j}{e_j}.$$

Under H_0, for large n, $\Lambda \approx \chi^2(k - 1 - p)$.

Derivation. $L(\theta) \propto \prod \theta_j^{y_j}$. Unrestricted maximum at $\hat{\theta}_j = y_j / n$; under H_0 at $\theta_j(\hat{\alpha}) = e_j / n$. So $\Lambda = 2 \sum y_j \big[\log(y_j / n) - \log(e_j / n) \big] = 2 \sum y_j \log(y_j / e_j)$. The full model has $k - 1$ free parameters and H_0 has p, so by the general LRT result the degrees of freedom are $(k - 1) - p$. □

Definition 7.3 (Pearson's chi-squared statistic)**.**

$$D = \sum_{j=1}^{k} \frac{(y_j - e_j)^2}{e_j},$$

which is a quadratic approximation to Λ (expand $y\log(y/e)$ around $y = e$) and has the same approximate $\chi^2(k-1-p)$ distribution under H_0.

Degrees of freedom: (number of categories -1) $-$ (number of parameters estimated under H_0). **Rule of thumb:** the χ^2 approximation is adequate if all $e_j \geq 5$; otherwise combine adjacent categories. A term $y_j \log(y_j/e_j)$ with $y_j = 0$ is taken as 0.

Example 7.4 (Is the die fair? $(p = 0)$)**.** A die rolled $n = 120$ times gives

face	1	2	3	4	5	6
observed y_j	14	23	17	25	16	25
expected e_j	20	20	20	20	20	20

$H_0 : \theta_j = 1/6$ for all j, fully specified, so df $= 6 - 1 - 0 = 5$.

$$D = \frac{36 + 9 + 9 + 25 + 16 + 25}{20} = \frac{120}{20} = 6.00, \qquad \Lambda = 2\sum y_j \log \frac{y_j}{20} = 6.09.$$

For $W \sim \chi^2(5)$, $\mathrm{P}(W \leq 9.236) = 0.90$, so $p > 0.10$ (software: $p \approx 0.30$ for both). No evidence that the die is unfair.

Example 7.5 (Poisson goodness of fit $(p = 1)$)**.** The car counts of Chapter 2 in $n = 200$ intervals had observed frequencies $38, 80, 50, 20, 12$ for $0, 1, 2, 3, \geq 4$ cars, $\hat{\theta} = 1.455$ and expected frequencies $46.68, 67.92, 49.41, 23.96, 12.02$ (all ≥ 5). The contributions to D are

$$\frac{(38 - 46.68)^2}{46.68} = 1.614, \quad 2.149, \quad 0.007, \quad 0.656, \quad 0.000,$$

so $D = 4.43$; similarly $\Lambda = 2\sum y_j \log(y_j/e_j) = 4.46$. Degrees of freedom: $5 - 1 - 1 = 3$ (one parameter θ was estimated). For $W \sim \chi^2(3)$, $\mathrm{P}(W \leq 6.251) = 0.90$, so $p > 0.10$ (software $p \approx 0.22$). No evidence against the Poisson model.

7.3 Two-way tables and tests of independence

Each of n units is classified by two categorical variates A (with a levels) and B (with b levels), giving counts y_{ij} in an $a \times b$ *contingency table*, with row totals $r_i = \sum_j y_{ij}$ and column totals $c_j = \sum_i y_{ij}$.

Model: multinomial with cell probabilities θ_{ij}, $ab - 1$ free parameters. The hypothesis of *independence* of A and B is

$$H_0 : \theta_{ij} = \alpha_i \beta_j \quad \text{for all } i, j,$$

with $\sum \alpha_i = \sum \beta_j = 1$, i.e. $(a-1) + (b-1)$ free parameters.

Theorem 7.6 (Test of independence)**.** *Under H_0 the MLEs are $\hat{\alpha}_i = r_i/n$, $\hat{\beta}_j = c_j/n$, so the expected frequencies are*

$$e_{ij} = n\hat{\alpha}_i\hat{\beta}_j = \frac{r_i c_j}{n}.$$

The statistics $\Lambda = 2\sum_{i,j} y_{ij}\log(y_{ij}/e_{ij})$ and $D = \sum_{i,j}(y_{ij} - e_{ij})^2/e_{ij}$ are approximately $\chi^2((a-1)(b-1))$ under H_0, since $(ab-1) - (a-1) - (b-1) = (a-1)(b-1)$.

Remark. The same computation tests *homogeneity*: if a separate random samples are taken (e.g. fixed numbers of men and women) and each classified by B, the hypothesis that the B-distribution is the same in all a populations leads to identical e_{ij}, statistic and degrees of freedom.

Example 7.7 (2×3 table)**.** A random sample of 250 students classified by faculty and preferred course format:

	online	hybrid	in person	total
Math	60 (48)	45 (48)	15 (24)	120
Arts	40 (52)	55 (52)	35 (26)	130
total	100	100	50	250

Expected frequencies in parentheses, e.g. $e_{11} = 120(100)/250 = 48$, $e_{13} = 120(50)/250 = 24$. The contributions to D are

$$\begin{array}{ccc} 3.000 & 0.188 & 3.375 \\ 2.769 & 0.173 & 3.115 \end{array}$$

so $D = 12.62$ (and $\Lambda = 12.86$), with $(2-1)(3-1) = 2$ degrees of freedom. For $W \sim \chi^2(2)$, $\mathrm{P}(W \le 9.210) = 0.99$, so $p < 0.01$ (software: $p \approx 0.002$). Strong evidence that format preference is associated with faculty: Math students prefer online more, Arts students prefer in-person more than independence predicts.

Example 7.8 (2×2 table)**.** Of 50 patients on treatment A, 30 improved; of 50 on treatment B, 18 improved. Totals: 48 improved, 52 did not. Expected: $e = 50(48)/100 = 24$ improved and 26 not, in each group. Then

$$D = \frac{6^2}{24} + \frac{6^2}{26} + \frac{6^2}{24} + \frac{6^2}{26} = 1.5 + 1.385 + 1.5 + 1.385 = 5.77,$$

on 1 degree of freedom; $p = \mathrm{P}(W \ge 5.77) = \mathrm{P}(|Z| \ge 2.40) \approx 0.016$ ($\Lambda = 5.83$ gives $p \approx 0.016$ too). Evidence that improvement rates differ. (For a 2×2 table this is equivalent to the two-sided test comparing two binomial proportions.)

Common pitfalls

- Use *counts*, never percentages, in D and Λ.
- Subtract one degree of freedom for each parameter estimated from the data under H_0.
- Check $e_j \ge 5$; merge sparse categories before testing.
- A significant test of independence shows association, not causation.
- These tests are always "one-tailed" in the sense that only large D counts as evidence against H_0.

Summary

- Multinomial MLE: $\hat{\theta}_j = y_j/n$.
- Goodness of fit: $\Lambda = 2\sum y_j \log(y_j/e_j)$ or $D = \sum(y_j - e_j)^2/e_j$, approximately $\chi^2(k-1-p)$.
- Independence in an $a \times b$ table: $e_{ij} = r_i c_j/n$, $\chi^2((a-1)(b-1))$.

Chapter 8

Causal Relationships

8.1 Association versus causation

Definition 8.1. Variates x and y are *associated* if the distribution of y differs for different values of x (e.g. a non-zero correlation, a non-zero regression slope, or dependence in a two-way table). We say x *causes* y if, other things being held fixed, changing the value of x for a unit changes the (distribution of the) value of y for that unit.

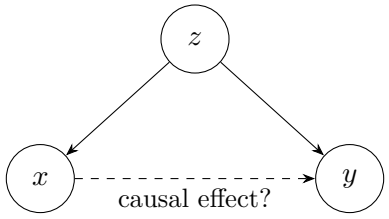
Association is about what we *observe*; causation is about what would happen if we *intervened*. All the tests of Chapters 5–7 detect association. Whether we may interpret it causally depends on how the data were collected.

Example 8.2 (Associations that are not causal)**.**

- Cities with more firefighters have more fire damage (both are driven by city size and number of fires).
- Ice cream sales and drownings are positively correlated (both increase in hot weather).
- Students who use the tutoring centre have lower grades (weaker students are more likely to seek help: *reverse causation* / selection).

8.2 Confounding

Definition 8.3 (Confounding variate)**.** A variate z is a *confounder* for the relationship between x and y if z is associated with x and also affects y. Then an observed association between x and y may be partly or wholly due to z, and the effects of x and z cannot be separated from the data.



Strategies for dealing with confounding:

- **Randomization** (in experiments): breaks any link between x and every possible confounder, known or unknown.

- **Stratification/matching**: compare x-groups within levels of a known confounder z (e.g. within age groups); pairing (Chapter 6) is an example.
- **Including z in the model** (multiple regression, beyond Stat 231).

Only randomization handles *unknown* confounders.

8.3 Experimental versus observational studies

Definition 8.4. In an *experimental study* the investigator controls (assigns) the value of the explanatory variate x for each unit. In an *observational study* the values of x are not controlled; they are just recorded as they occur.

Principles of a good experiment

- **Control/comparison**: compare the treatment with a control group (e.g. placebo), keeping other conditions the same.
- **Randomization**: assign treatments to units at random, so that the groups are alike on average in all other respects; then a difference in y can be attributed to x (or chance, which the p-value quantifies).
- **Replication**: use enough units to make the sample error small.
- **Blinding**: if possible, neither units (single-blind) nor evaluators (double-blind) know which treatment was received, avoiding placebo effects and measurement bias.

For many important questions (smoking and cancer, pollution and asthma) experiments are unethical or impossible, so we must rely on observational studies: *cohort* studies (follow exposed and unexposed groups forward in time) and *case-control* studies (compare past exposure of people with and without the disease). Such studies can be very informative but are always open to confounding.

8.4 Simpson's paradox

Definition 8.5. *Simpson's paradox* occurs when an association between two variates in the combined data has the opposite direction from the association within every subgroup defined by a third variate. It is an extreme form of confounding.

Example 8.6 (Kidney stone treatments)**.** Success rates of two treatments in an observational study:

	small stones	large stones	all patients
Treatment A	$81/87 = 93\%$	$192/263 = 73\%$	$273/350 = 78\%$
Treatment B	$234/270 = 87\%$	$55/80 = 69\%$	$289/350 = 83\%$

A is better for small stones and better for large stones, yet B looks better overall. The explanation: stone size is a confounder. Large stones are harder to treat (lower success for both treatments), and doctors gave A mostly to large-stone patients (263 of 350) and B mostly to small-stone patients (270 of 350). The overall comparison mixes the treatment effect with the severity effect. The

stratified comparison (within stone size) is the relevant one, and a randomized trial would avoid the problem.

Remark. Algebraically, the overall rate is a weighted average of the subgroup rates, with weights equal to the fraction of each group's patients in each subgroup: $0.78 = \frac{87}{350}(0.931) + \frac{263}{350}(0.730)$ and $0.826 = \frac{270}{350}(0.867) + \frac{80}{350}(0.688)$. Different weights can reverse the order.

8.5 Establishing causation from observational data: Hill's criteria

When experiments are impossible, the epidemiologist Austin Bradford Hill (1965) proposed considerations that strengthen the case for a causal interpretation of an association. Stat 231 emphasizes:

1. **Strength**: the association is large (e.g. heavy smokers have many times the lung cancer rate of non-smokers); a strong association is harder to explain away by confounding.
2. **Consistency**: the association is found repeatedly in different studies, populations, designs and by different investigators.
3. **Specificity**: the cause leads to a specific effect rather than many unrelated ones.
4. **Temporality**: the cause precedes the effect.
5. **Dose–response (biological gradient)**: more exposure leads to more effect.
6. **Plausibility**: there is a credible mechanism (e.g. carcinogens in smoke).
7. **Coherence**: the causal interpretation does not conflict with known facts about the disease.
8. **Experimental evidence**: e.g. animal experiments, or the effect diminishes when exposure is removed (people who quit smoking have reduced risk).
9. **Analogy**: similar causes are known to have similar effects.

No single criterion is decisive (except temporality, which is necessary), but together they built the overwhelming case that smoking causes lung cancer without a randomized trial.

Common pitfalls

- A small p-value says an association is unlikely to be due to chance alone; it says nothing about whether it is causal.
- Random *sampling* lets you generalize to a population; random *assignment* lets you conclude causation. Neither implies the other.
- Always ask "what else differs between the groups?"

Summary

- Association $\neq$ causation; confounders, reverse causation and selection can produce associations.
- Randomized controlled experiments are the gold standard for causal conclusions because randomization balances all confounders on average.

- Simpson's paradox: aggregating over a confounder can reverse an association.
- For observational data, Hill's criteria (strength, consistency, temporality, dose–response, plausibility, ...) support but never prove causation.

Chapter 9

Summary Tables and References

This chapter collects the formulas used throughout the course for quick reference.

9.1 Models and maximum likelihood estimates

Model	MLE	Approx. pivotal / standard error	Notes
Binomial(n, θ)	y/n	$\sqrt{\hat{\theta}(1-\hat{\theta})/n}$	CLT
Poisson(θ), n obs.	$\bar{y}$	$\sqrt{\hat{\theta}/n}$	CLT
Exponential(θ), n obs.	$\bar{y}$	$2\sum Y_i/\theta \sim \chi^2(2n)$	exact
G(μ, σ)	$\bar{y}$, $\frac{1}{n}\sum(y_i-\bar{y})^2$	see below	exact
Multinomial	y_j/n	—	GOF tests

9.2 Pivotal quantities for Gaussian models

Setting	Pivotal quantity	Distribution
mean, σ known	$\dfrac{\bar{Y}-\mu}{\sigma/\sqrt{n}}$	G$(0,1)$
mean, σ unknown	$\dfrac{\bar{Y}-\mu}{S/\sqrt{n}}$	$t(n-1)$
variance	$\dfrac{(n-1)S^2}{\sigma^2}$	$\chi^2(n-1)$
regression slope	$\dfrac{\tilde{\beta}-\beta}{S_e/\sqrt{S_{xx}}}$	$t(n-2)$
mean response at x	$\dfrac{\tilde{\mu}(x)-\mu(x)}{S_e\sqrt{\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}}}$	$t(n-2)$
new response at x	$\dfrac{Y-\tilde{\mu}(x)}{S_e\sqrt{1+\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}}}$	$t(n-2)$
two samples, common σ	$\dfrac{\bar{Y}_1-\bar{Y}_2-(\mu_1-\mu_2)}{S_p\sqrt{1/n_1+1/n_2}}$	$t(n_1+n_2-2)$
paired differences	$\dfrac{\bar{D}-\mu_d}{S_d/\sqrt{n}}$	$t(n-1)$

9.3 Likelihood-based results

- Relative likelihood $R(\theta) = L(\theta)/L(\hat{\theta})$; $100p\%$ likelihood interval $\{\theta : R(\theta) \geq p\}$.
- $\Lambda(\theta) = -2\log R(\theta) \approx \chi^2(1)$; 15% LI $\approx$ 95% CI, 10% LI $\approx$ 96.8% CI, 1% LI $\approx$ 99.8% CI.
- LRT of $H_0 : \theta = \theta_0$: $p \approx \mathrm{P}(W \geq \lambda) = 2[1 - \Phi(\sqrt{\lambda})]$, $W \sim \chi^2(1)$.
- Goodness of fit: $\Lambda = 2\sum y_j \log(y_j/e_j)$, $D = \sum(y_j - e_j)^2/e_j$, $\chi^2(k-1-p)$; independence: $e_{ij} = r_i c_j / n$, $\chi^2((a-1)(b-1))$.

9.4 Frequently used table values

Standard normal: $\mathrm{P}(Z \leq z)$ for $z = 1.282, 1.645, 1.960, 2.326, 2.576$ is $0.90, 0.95, 0.975, 0.99, 0.995$.

Student t: values a with $\mathrm{P}(T \leq a) = 0.975$.

df	7	8	9	10	14	19	29	60	∞
a	2.365	2.306	2.262	2.228	2.145	2.093	2.045	2.000	1.960

Chi-squared: quantiles c with $\mathrm{P}(W \leq c) = q$.

df	$q = 0.025$	$q = 0.05$	$q = 0.10$	$q = 0.90$	$q = 0.95$	$q = 0.975$
1	0.001	0.004	0.016	2.706	3.841	5.024
2	0.051	0.103	0.211	4.605	5.991	7.378
3	0.216	0.352	0.584	6.251	7.815	9.348
4	0.484	0.711	1.064	7.779	9.488	11.143
5	0.831	1.145	1.610	9.236	11.070	12.833
9	2.700	3.325	4.168	14.684	16.919	19.023
16	6.908	7.962	9.312	23.542	26.296	28.845

9.5 Workflow reminders

1. Problem and Plan: identify target and study populations, variates, sampling protocol; anticipate study, sample and measurement error.
2. Plot the data; choose a model; estimate by maximum likelihood; check the fit.
3. Report estimates with confidence or likelihood intervals; test hypotheses with p-values interpreted on the evidence scale.
4. Draw conclusions in context, distinguishing association from causation.

References

- University of Waterloo, Department of Statistics and Actuarial Science, *Stat 231 Course Notes* (editions used 2016–2021).
- J. G. Kalbfleisch, *Probability and Statistical Inference*, Vol. 2, Springer.
- A. B. Hill, "The environment and disease: association or causation?", *Proceedings of the Royal Society of Medicine*, 1965.